16. Ans: C

$$\frac{1}{2}m(c_{\text{rms}})^2 = \frac{3}{2}kT$$

$$m(c_{\text{rms}})^2 \propto T$$

$$\frac{m_{He}(c_{\text{rms}})^2{}_{He}}{m_{Ne}(c_{\text{rms}})^2{}_{Ne}} = 1 \quad \text{(thermal equilibrium)}$$

$$\frac{(c_{\text{rms}})^2{}_{He}}{(c_{\text{rms}})^2{}_{Ne}} = \frac{m_{Ne}}{m_{He}}$$

$$\frac{c_{\text{rms}He}}{c_{\text{rms}Ne}} = \sqrt{\frac{m_{Ne}}{m_{He}}}$$

$$= \sqrt{2.5}$$

$$= 1.6$$

17. Ans: A